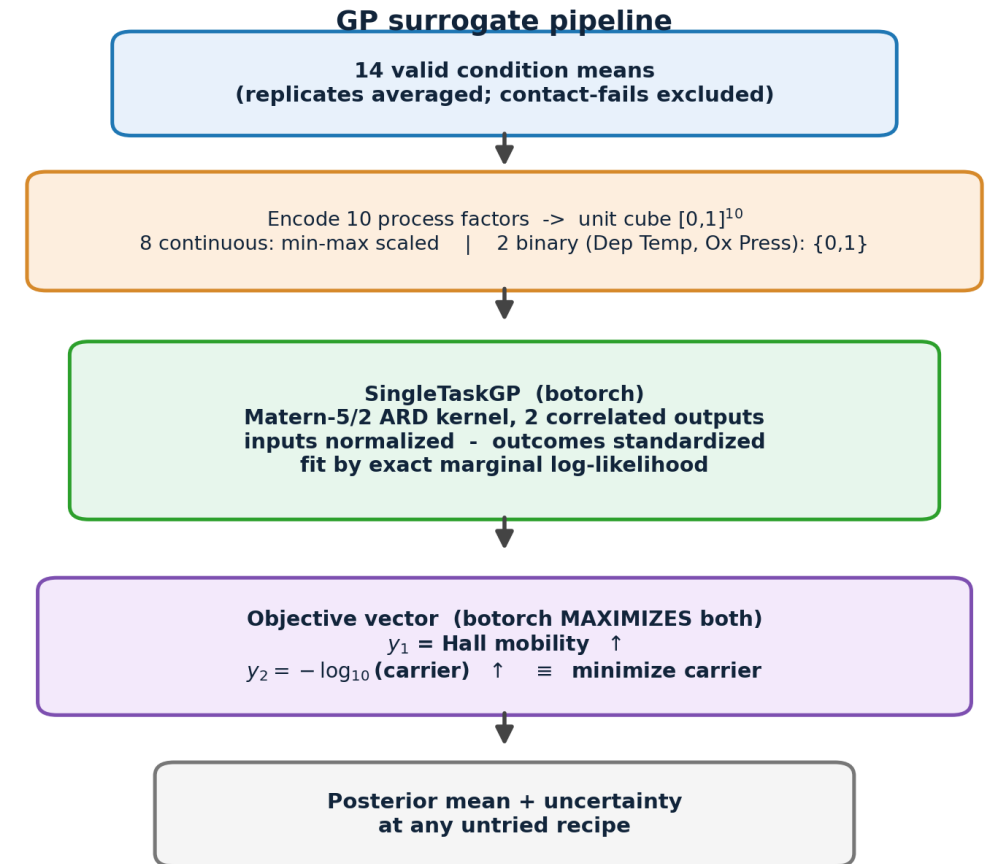


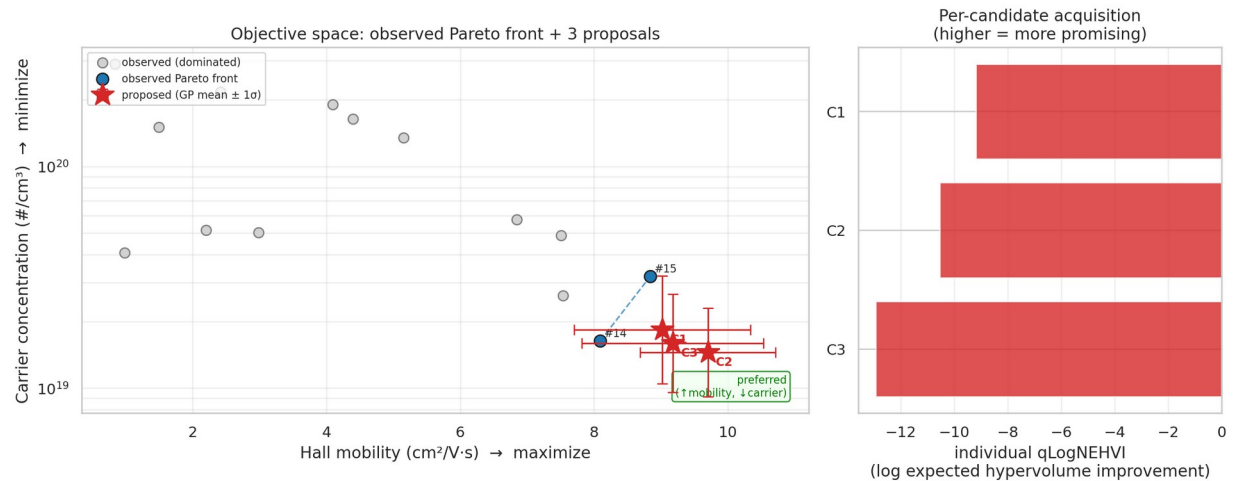
Surrogate model: multi-output Gaussian Process

- Why a GP? Linear models (§5) were not predictive at $n=14$; a GP is built for small, noisy, high-dimensional data and returns calibrated uncertainty.
- Training data: 14 valid condition means — replicates averaged, contact-fail runs excluded (failures are not modeled).
- **10 process factors encoded to the unit cube $[0,1]^{10}$:**
 - 8 continuous factors: min-max scaled to their bounds
 - 2 binary factors (Dep Temp, Ox Press): mapped to $\{0, 1\}$
- **Model: botorch SingleTaskGP — 2 correlated outputs, Matérn-5/2 ARD kernel, inputs normalized, outcomes standardized; fit by exact marginal log-likelihood.**
- **Objectives (botorch maximizes): $y_1 = \text{mobility} \uparrow$; $y_2 = -\log_{10}(\text{carrier}) \uparrow \equiv \text{minimize carrier}$. Carrier in $\log_{10}$ (spans > 1 decade).**
- Output: posterior mean + uncertainty at any untried recipe $\rightarrow$ drives the acquisition search (next slide).



Acquisition: qLogNEHVI (batch multi-objective)

- Hypervolume = the scalar quality of the Pareto front: the area/volume it dominates above a reference point. Observed HV = 11.89.
- qLogNoisyExpectedHypervolumeImprovement scores a recipe by its expected increase in that hypervolume.
 - Noisy: integrates over GP posterior → robust to observation noise
 - Log: log-space evaluation → numerically stable optimization
 - $q = 3$: optimizes a joint batch of 3 (diversity built in); joint score -7.84
- **Mixed optimization (optimize_acqf_mixed):**
 - enumerate the 4 binary combos (Dep Temp × Ox Press) as fixed features
 - optimize the 8 continuous knobs on $[0,1]$ — 20 restarts, 256 QMC samples
- **Failure-aware search bounds: Vac Ann ≤ 550 °C (baked into encoding); Ox Temp ≤ 680 °C for proposals only — GP still trained on the full range.**

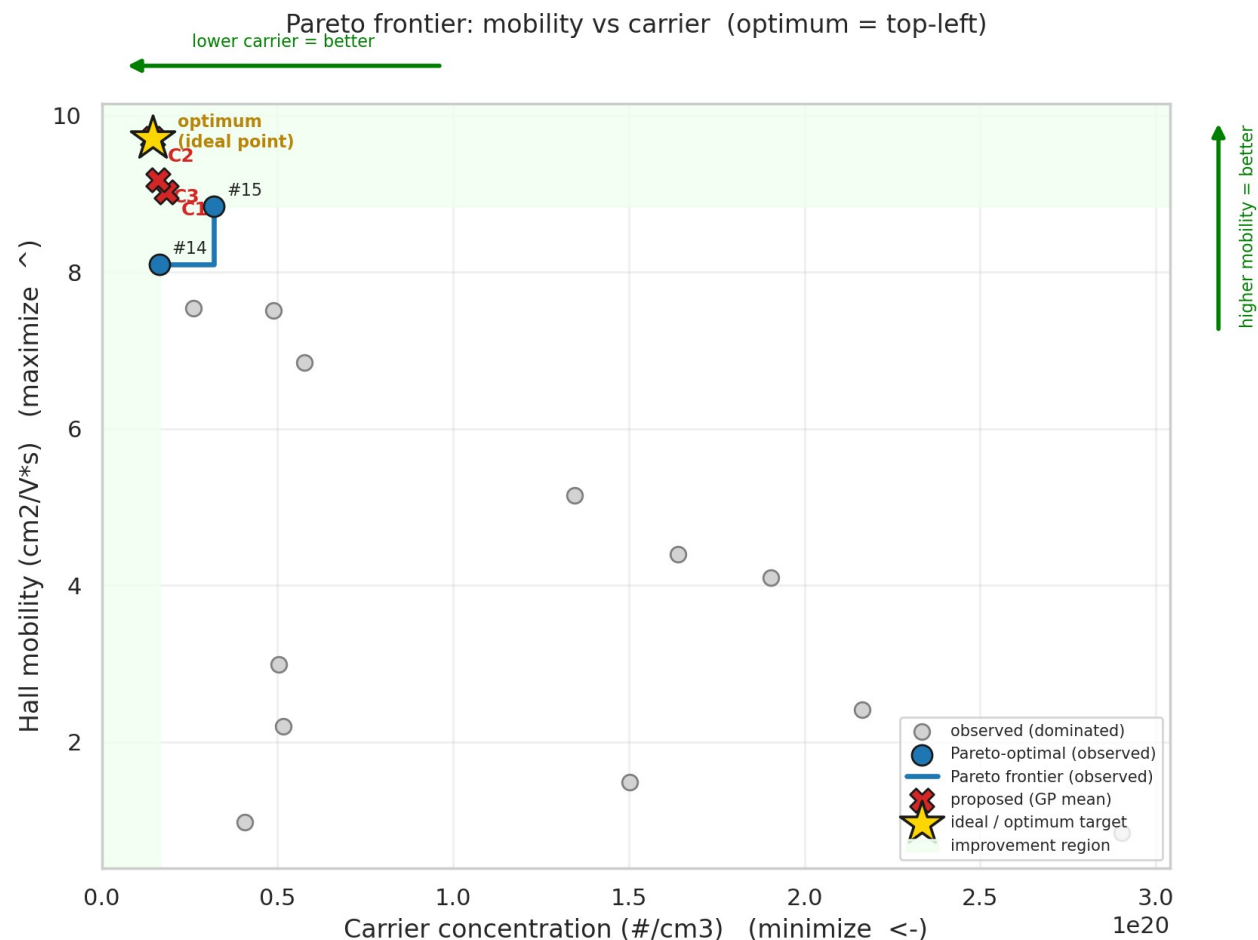


Objective plane: observed Pareto front + 3 proposals (GP mean $\pm 1\sigma$); right bar = per-candidate qLogNEHVI

Output: 3 proposed next experiments

- No single winner — the optimizer returns a Pareto set. The 3 candidates are the points qLogNEHVI expects to expand the front the most.
- Optimum corner = top-left: high mobility, low carrier (gold ideal point).**
- Ranked by individual qLogNEHVI → C1 is the most promising single recipe.
- Predictions carry $\pm 1\sigma$ uncertainty; proposals push beyond the current front toward the ideal point.

Cand.	Pred. mobility (cm ² /V·s)	Pred. carrier (#/cm ³)	qLogNEHVI (rank)
C1	9.02 ± 1.32	1.84e19	−9.19 (1)
C2	9.70 ± 1.01	1.45e19	−10.55 (2)
C3	9.18 ± 1.36	1.60e19	−12.94 (3)



Pareto frontier (linear carrier axis): optimum = top-left; gold ★ = ideal point; shaded = improvement region